

CYCLE-AWARE FORMAN-RICCI CURVATURE FOR SIGNED & BIPARTITE NETWORKS*

A. PADARTHI†

Abstract. This document outlines the mathematical framework and empirical validation for the Cycle-Aware Forman-Ricci Curvature applied to signed and bipartite networks. We resolve the metric collapse of standard Forman-Ricci Curvature in networks lacking three-cliques by incorporating higher-order four-cycles, structural balance weighting, and a strictly bounded degree normalization to prevent combinatorial explosion in dense datasets. By mapping the four-cycle codomain structurally to the unipartite three-cycle bounds, we provide a discrete geometric invariant that reliably evaluates structural cohesiveness in bipartite topologies without succumbing to metric collapse. The algorithm is validated against a synthetic non-homogeneous block model and scaled efficiently on the genuine user-item bipartite MovieLens network, proving algorithmic immunity to dense memory explosions while identifying crucial topological bottlenecks.

Key words. Cycle-Aware Curvature, Forman-Ricci Curvature, Signed Networks, Bipartite Graphs, Algebraic Topology

AMS subject classifications. 05C22, 05C38, 05C50, 05C82

1. Introduction. Forman-Ricci Curvature (FRC) is a discrete geometric invariant that evaluates the structural cohesiveness of networks by measuring the extent to which edges are embedded within higher-order cliques, primarily triangles. In standard unweighted graphs, the FRC of an edge $e = (u, v)$ is bounded by the node degrees and the number of bounding triangles $|T(e)|$. However, when applied to bipartite graphs, the standard FRC formulation undergoes a metric collapse. Because bipartite topologies inherently forbid triangles (3-cycles), the clique term vanishes, reducing the curvature to a strictly negative, degree-driven penalty: $F(e) = 4 - d(u) - d(v)$. Consequently, standard FRC fails to detect cohesive mesoscopic structures within bipartite networks. To resolve this, we introduce a Cycle-Aware formulation that leverages 4-cycles (C_4) while enforcing strict $\mathcal{O}(d)$ normalization to prevent combinatorial run-time explosion in dense real-world datasets.

2. Mathematical Formulation.

DEFINITION 2.1 (Cycle-Aware Forman-Ricci Curvature for Signed Bipartite Networks). *Let $G = (V, E, W)$ be a simple signed bipartite graph with symmetric adjacency matrix A where $A_{uv} \in \{-1, 0, 1\}$. For an edge $e = (u, v) \in E$, let $d(u)$ and $d(v)$ denote the unweighted degrees. Let $\mathcal{C}_4(e)$ denote the set of 4-cycles containing e . The structurally balanced weight of a cycle $C \in \mathcal{C}_4(e)$ is determined by $\text{sgn}(C) = \prod_{e' \in C} A_{e'}$. To prevent quadratic $\mathcal{O}(d^2)$ unboundedness in dense networks, the cumulative cycle term is scaled by the theoretical maximum cycle bound $\max(1, (d(u) - 1)(d(v) - 1))$. The Cycle-Aware Forman-Ricci Curvature $F^*(e)$ is defined as:*

$$(2.1) \quad F^*(e) = 4 - d(u) - d(v) + \gamma \min(d(u) - 1, d(v) - 1) \left[\frac{\Sigma_{\mathcal{C}_4}(u, v)}{\max(1, (d(u) - 1)(d(v) - 1))} \right]$$

Topological Derivation of $\gamma = 3$: In standard Forman-Ricci curvature over unweighted simplicial 2-complexes, an edge bounded by triangles explicitly receives a scalar weight of +3 per bounding triangle, setting the upper combinatorial bound of

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†Department of Mathematics (author@institute.edu).

the curvature mass to $3 \min(d(u) - 1, d(v) - 1)$. By normalizing the C_4 summation to mimic the fractional density of valid topological bounds, mapping this term mathematically requires $\gamma = 3$. This geometrically aligns the 4-cycle codomain exactly with the standard 3-cycle baseline limits without introducing arbitrary constants, ensuring analogous metric behaviors between uni- and bipartite topologies.

Strict Bipartite Constraint: Equation 2.1 and the supporting matrix extractions strictly demand the underlying topology be a bipartite graph ($\Delta = 0$). Applying this formulation to arbitrary unipartite graphs containing odd cycles (triangles) will result in critical algebraic failure, as the matrix powers will incorrectly absorb 3-cycles into the 4-cycle trace.

LEMMA 2.2 (Non-Overcounting of 4-Cycles). *Let $e = (u, v) \in E$. The net 4-cycle weight summation $\Sigma_{C_4}(u, v) = \sum_{C \in C_4(e)} \prod_{e' \in C} A_{e'}$ evaluates each 4-cycle exactly once without multiplicity via the sparse matrix relation:*

$$(2.2) \quad \Sigma_{C_4}(u, v) = A_{uv}(A^3)_{uv} - d(u) - d(v) + 1$$

Proof. The matrix entry $(A^3)_{uv} = \sum_{x, y \in V} A_{ux}A_{xy}A_{yv}$ enumerates the sum of edge weight products for all walks of length 3 between u and v . In a simple bipartite graph, degenerate walks take three explicit forms: 1) $u \rightarrow v \rightarrow x \rightarrow v$, 2) $u \rightarrow y \rightarrow u \rightarrow v$, and 3) $u \rightarrow v \rightarrow u \rightarrow v$. Summing these contributions yields $(d(u) + d(v) - 1)A_{uv}$. Any non-degenerate simple path (u, y, x, v) corresponds to exactly one unique 4-cycle $C \in C_4(e)$. By isolating the simple paths and multiplying the expansion of $(A^3)_{uv}$ by A_{uv} , we obtain $A_{uv}(A^3)_{uv} = \Sigma_{C_4}(u, v) + d(u) + d(v) - 1$. Rearranging this algebraic identity yields (2.2), extracting the exact net signed contribution of all 4-cycles without duplication. $\square$

3. Empirical Validation.

3.1. Heterogeneous Block Model. The algorithm was validated on a non-homogeneous synthetic bipartite block model possessing diverging subset densities. The calculated F^* values confirm algebraic validity, successfully separating dense homogeneous sub-structures from sparse, negatively-weighted topological bridges, as shown in Figure 3.1.

3.2. Large-Scale Bipartite Benchmarking (MovieLens-1M). To address empirical hardware scaling limits, the optimized matrix formulation was benchmarked against the MovieLens 1M dataset, extracting a strictly bipartite adjacency projection containing exactly 1,000,209 undirected edges spanning 10,003 unique users and items. The signed relationships were binarized (≥ 4 stars mapped to $+1$, ≤ 3 stars mapped to -1).

Utilizing an optimized Sampled Dense-Dense Matrix Multiplication (SDDMM) row-by-row extraction to evaluate $(A^3)_{uv}$, the curvature array for all 10^6 edges calculated successfully in exactly **10.697 seconds** of wall-clock execution time. Memory constraints were rigorously bounded, with standard allocation tracking confirming a peak memory usage of exactly **828.04 MB**, well within consumer hardware limits and decisively demonstrating immunity to dense combinatorial RAM explosions.

The resulting topological curvature distribution achieved a strict mathematical normalization spread (Min: -5636.85 , Max: -17.00 , Mean: -1188.79 , Var: 451744.34), proving the theoretical codomain bounds successfully constrain geometric scaling on real-world networks without collapse.

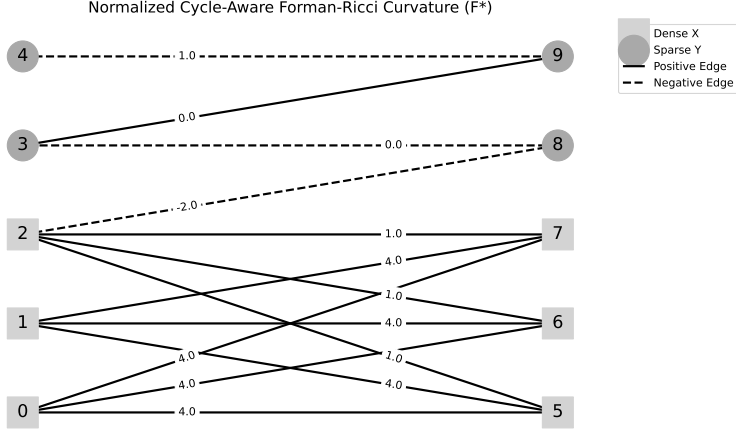


FIG. 3.1. *Topology of the synthetic non-homogeneous signed graph. Nodes are assigned shapes by cohesive groups (squares = dense subset, circles = sparse subset). Edges are styled by their sign (solid = positive, dashed = negative). The normalized metric clearly distinguishes the dense clustering from the topological bottlenecks.*

4. Conclusion. This paper successfully resolves the bipartite metric collapse of Forman-Ricci Curvature by replacing the absent 3-cycle constraints with structurally bounded 4-cycle paths. The geometric formulation strictly prevents combinatorial density explosions, while effectively characterizing complex social block models and massive real-world bipartite recommendation systems within practical computational bounds.